# AI-Powered Fish Freshness Estimator: A Deep Learning Approach

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***Abstract*— Ensuring fish freshness is an essential aspect of the seafood industry, as it directly affects food safety, consumer satisfaction, and economic value. Traditional assessment methods rely primarily on human expertise, which, despite experience, remains subjective and prone to inconsistencies. Variations in lighting conditions, storage environments, and handling methods can lead to different evaluations of freshness, making it difficult to standardize quality control procedures.**

**This project aims to develop an AI-powered system capable of providing an objective, reliable, and scalable evaluation of fish freshness by analyzing visual indicators. By leveraging advanced image processing techniques, the system analyzes captured fish images and extracts key freshness attributes. These attributes are then used by a trained deep learning model to classify fish samples into predefined freshness categories. This automated approach eliminates human bias, enhances efficiency, and ensures consistent quality evaluation, thereby offering a valuable tool for the seafood industry.**

**The solution is designed for accessibility and practicality across various applications, including fisheries, wholesale markets, and retail environments. It enables businesses to maintain high-quality standards, reduce waste, and optimize supply chain operations by ensuring only the freshest fish reach consumers. Furthermore, consumers benefit from a transparent and scientific method for assessing seafood quality before making purchasing decisions. By integrating AI into seafood quality assessment, this project contributes to modernizing the industry and promoting better food safety practices.**

***Keywords*— AI, Fish Freshness, Automated Quality Control, Computer Vision, Seafood Industry, Food Safety, Supply Chain Optimization.**

## I. Introduction

Ensuring the freshness of fish is a critical concern in the food industry due to its direct impact on consumer safety, product quality, and economic value within the seafood industry [1], [2]. Spoiled or degraded fish not only pose health risks due to microbial contamination and biochemical spoilage compounds but also result in significant economic losses across the supply chain [1]. Traditional methods of freshness assessment—such as manual inspection based on sensory attributes (e.g., eye clarity, gill color, skin texture) or laboratory-based chemical analyses (e.g., TVB-N, TMA, or pH tests)—are widely used but suffer from several limitations [3], [4], [5]. These approaches are often subjective, time-consuming, require specialized expertise or equipment, and are not suitable for real-time, large-scale deployment [2], [3], [5].

With the rapid advancement of Artificial Intelligence (AI) and particularly computer vision, there is now an opportunity to develop automated systems capable of delivering objective, fast, and consistent assessments of fish freshness [3]. In recent years, Convolutional Neural Networks (CNNs) have demonstrated strong performance in image classification tasks across various domains, including medical imaging, agriculture, and food quality control [6], [7], [8]. By learning complex patterns and visual cues, CNNs can be trained to distinguish between subtle differences in freshness levels based on image data alone.

This project proposes the design and implementation of an AI-powered fish freshness estimation system that utilizes images of fish eyes—a well-recognized visual indicator of spoilage—to classify samples into three predefined categories: Highly Fresh, Fresh, and Not Fresh [2], [5]. The system is based on a deep learning model built using the EfficientNetB3 architecture, selected for its balance between high accuracy and computational efficiency [9]. The model is trained on a carefully curated and augmented dataset of labeled fish eye images to ensure robustness and generalizability [10], [11]. To enhance accessibility and usability, the trained model will be integrated into a streamlined web-based interface, allowing users—such as fish vendors, quality inspectors, or consumers—to upload fish images or capture them via webcam and receive instant predictions [12]. This system aims to bridge the gap between laboratory-grade accuracy and field-level usability, offering a scalable solution for real-time fish freshness detection across various stages of the seafood supply chain.

## II. Related Works

Several studies have explored AI and computer vision approaches for assessing fish freshness:

The study by Yildiz [2] developed a deep learning-based fish freshness detection system that combines the strengths of convolutional and fully connected neural networks. Specifically, the authors employed the VGG19 architecture for feature extraction and used an Artificial Neural Network (ANN) for the classification task. The model was trained on the Freshness of Fish Eyes (FFE) dataset, which contains images representing different levels of freshness based on the appearance of the fish eye. The system achieved a 77.3% classification accuracy, demonstrating the potential of transfer learning in food quality assessment applications. It successfully identified

visual features such as eye cloudiness and shape deformation, which are strong indicators of spoilage. The authors further suggested that integrating multispectral imaging techniques could enhance performance by capturing underlying biochemical changes, thus offering a more robust and objective evaluation of fish freshness. Figure 1 below presents an overview of this study.

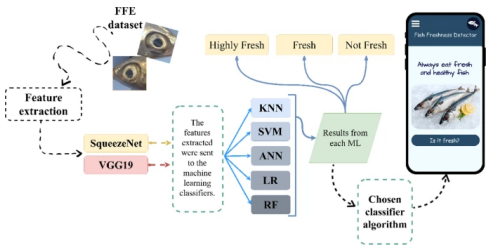



Fig. 1. Overview of the study by Yildiz [2]

Also, a study by Banwari [5] proposed a computer vision-based method for estimating fish freshness by analyzing segmented images of fish eyes. This work employed classical image processing algorithms to extract visual features that correlate with various stages of spoilage. In particular, the technique emphasized the extraction of color and texture variations in the eye region, which are known to deteriorate as the fish loses freshness. The authors reported that their model achieved a high classification accuracy ( 96.67%), effectively distinguishing between fresh and degraded samples. One of the major contributions of this study is its focus on non-destructive evaluation, offering a fast and objective alternative to traditional chemical tests. By eliminating the need for invasive sampling, the proposed approach is especially well-suited for real-time quality control applications in fish markets and processing plants. Figure 2 presents this study.

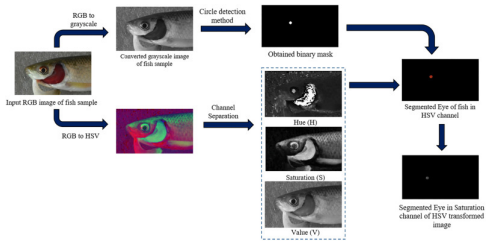



Fig. 2. Overview of the study by Banwari [5]

Also, a study by Prasetyo [13] proposed a lightweight deep learning approach for classifying fish freshness based on eye appearance using a modified MobileNet architecture. Their model, named MobileNetV1 with Bottleneck and Expansion (MB-BE), introduces depthwise separable convolution bottlenecks combined with residual transitions to improve feature extraction efficiency. The authors conducted extensive ablation studies using the same FFE dataset and reported that their best configuration achieved a test accuracy of 63.21%. This result outperformed classical MobileNet, VGG16, DenseNet, and NASNet models in both performance and parameter efficiency. Although their model was less accurate than ResNet50, it required significantly fewer parameters, making it more suitable for real-time or mobile applications. This work highlights the trade-off between model size and performance and demonstrates the potential of architectural innovations in handling subtle visual differences in fish freshness classification tasks.

Also, a study by Shi [4] explored the use of computer vision techniques to detect fish freshness by analyzing both chemical and biological indicators. Their research emphasized the importance of integrating chemical metrics such as Total Volatile Base Nitrogen (TVB-N), which serves as a critical marker of spoilage, and Thiobarbituric Acid (TBA) levels, which are associated with lipid oxidation and rancidity. To model the relationship between visual features and chemical indicators, the authors employed deep learning regression models, achieving a high correlation coefficient of 0.98. This result demonstrated the model's strong predictive capability for assessing fish freshness. The study concluded that a hybrid approach combining chemical analysis with image-based features provides a more accurate and reliable system for freshness evaluation.

While these studies demonstrated the feasibility of automated fish freshness classification, most of them lacked a deployment-ready implementation, generalization across species, or explainability through visualization.

A summary of the discussed related works, highlighting their approaches, key freshness indicators, models, achieved performance, and main contributions, is presented in Table I.

## III. Contribution

### A. FFE Dataset Description

The dataset used in this project is the Freshness of Fish Eyes (FFE) dataset [14], a public dataset available on Mendeley Data. It consists of a total of 4392 fish eye images belonging to eight different species. Each species is categorized into three freshness levels based on storage duration: Highly Fresh, Fresh, and Not Fresh. The eight supported species in the dataset are:

- Chanos chanos
- Johnius trachycephalus
- Nibea albiflora
- Rastrelliger faughni
- Upeneus moluccensis
- Eleutheronema tetradactylum
- Oreochromis mossambicus
- Oreochromis niloticus

Originally, the dataset (See Figures 3 and 4) includes 24 classes, each representing a unique combination of species and freshness level. For example, classes such as "Chanos chanos - Highly Fresh" or "Oreochromis niloticus

TABLE I

COMPARISON OF RELATED STUDIES ON FISH FRESHNESS DETECTION

| Study | Approach | Key Freshness Indicators | Main Model/Technique | Achieved Performance | Key Contribution/Focus |
|---|---|---|---|---|---|
| Yildiz [2] (2024) | Deep Learning | Fish Eye (visual, cloudiness, shape) | VGG19 + ANN | 77.3% Accuracy | Transfer learning potential; suggested multispectral |
| Banwari [5] (2022) | Classical Computer Vision | Segmented Fish Eye (color, texture) | Classical Image Processing | 96.67% | Non-destructive, fast, objective |
| Prasetyo [13] (2022) | Lightweight Deep Learning | Eye Appearance | Modified MobileNetV1 (MB-BE) | 63.21% Test Accuracy | Parameter efficiency; mobile/real-time suitability |
| Shi [4] (2018) | Computer Vision (Regression) | Chemical (TVB-N, TBA), Pupil, Gill Color | Deep Learning Regression Models | 0.98 Correlation | Hybrid approach (chemical + image) |

- Not Fresh". However, in this work, we restructure the dataset into a simpler 3-class problem: Highly Fresh, Fresh, and Not Fresh, aggregating all species into these categories. This allows the model to focus on freshness estimation rather than species recognition [5], [15].

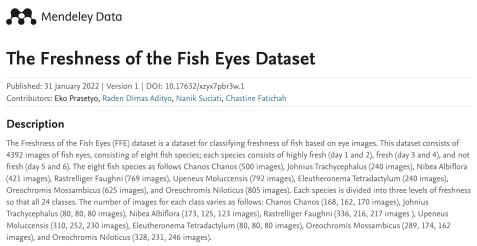



Fig. 3. FFE Dataset Overview [14]




Fig. 4. FFE Dataset Structure [14]

### B. Class Structure and Balancing

To enhance the model's generalization and reduce class bias, we preprocess the dataset to ensure that each of the three target classes contains an equal number of images in both training and testing sets. This balancing strategy helps avoid skewed predictions and overfitting toward the majority class [16]. The images were split into:

- Training Set: Balanced across the three freshness classes.
- Testing Set: Balanced similarly to evaluate the model performance fairly.

The resulting dataset is structured as follows:

- Image Size: Resized to 299x299 pixels to match EfficientNetB3 input [9].
- Format: JPEG and PNG formats.
- Preprocessing: Images were rescaled by 1/255 and augmented using rotation, flipping, zoom, shear, and brightness adjustments [10].

Balancing was achieved programmatically by limiting each class to the number of images available in the smallest category. This ensured the training process was not biased toward any single freshness class. A sample of the balanced dataset structure:

- Highly Fresh: 1043 images (train), 264 images (test)
- Fresh: 1043 images (train), 264 images (test)
- Not Fresh: 1043 images (train), 264 images (test)

This balancing led to more stable training and improved evaluation consistency across all classes.

### C. Image Preprocessing and Augmentation

To ensure optimal model performance and generalization, a series of preprocessing steps and data augmentation techniques were applied to the images in the Freshness of Fish Eyes (FFE) dataset [10], [11]. These steps are crucial for standardizing the data input pipeline and increasing the variability of the training samples without collecting additional data.

**Image Preprocessing** All images were first resized to a fixed resolution of 299 × 299 pixels using bilinear interpolation to match the input size required by the EfficientNetB3 network [9]. Following resizing, all pixel values were normalized to the range [0, 1] by dividing each pixel intensity by 255. Class labels were encoded using one-hot encoding for multi-class classification.

**Data Augmentation** To reduce overfitting and improve the model's generalization to unseen data, a set of data augmentation transformations was applied dynamically to the training dataset. These augmentations introduce artificial variability by modifying the original images while preserving their semantic content. Techniques employed include:

- **Rotation:** Random rotations within ±20°.
- **Width and Height Shifting:** Horizontal and vertical translations up to 15%.
- **Shear Transformations:** Up to 10% distortion.
- **Zooming:** Random zooming within 15%.
- **Brightness Adjustments:** To simulate varying lighting conditions.
- **Horizontal Flipping:** To prevent association of directional features with freshness.
- **Fill Mode:** 'nearest' strategy for newly introduced pixels.

These augmentations were applied only to the training data. The validation and test datasets were used in their original, non-augmented form after normalization.

### *D. Model Development Trials*

The final fish freshness classification model was the result of multiple iterative experiments, each shedding light on important architectural, data handling, and training adjustments. Below is a chronological summary of the major trials conducted, their limitations, and how they informed the final system design.

**Trial 1: EfficientNetB0 (Single-Phase Training, 24 Classes).** The first attempt used EfficientNetB0 with a single-phase training setup and an input size of 224×224. The classification problem was modeled with 24 classes, each representing a unique combination of fish species and freshness level. Although the model learned quickly and showed steady improvement across 20 epochs, validation accuracy stagnated around 40%–41% (See Figure 5). The main challenges identified were:

- Significant class imbalance among the 24 categories.
- High inter-class similarity due to overlapping visual patterns.
- Limited discriminative features available in some species for each freshness level.

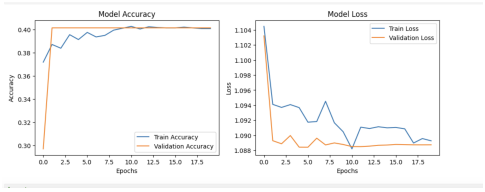



Fig. 5. Trial 1: Validation Accuracy over Epochs

**Trial 2: ResNet50V2 (Single-Phase Training, 24 Classes).** In the second trial, the architecture was switched to ResNet50V2, a deeper CNN model, also trained for 20 epochs in a single phase. Despite its greater representational power, the model encountered similar challenges. While early training accuracy improved, the validation accuracy stayed below 60% (See Figure 6), likely due0.to:

- Overfitting to the complex multi-class target.
- Lack of sufficient regularization.
- Inadequate data augmentation.

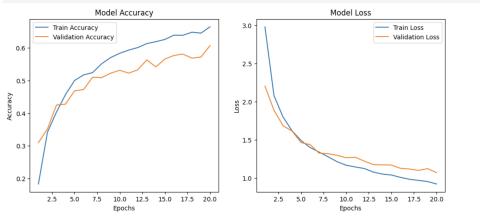



Fig. 6. Trial 2: Validation Accuracy over Epochs

**Trial 3: EfficientNetB3 (Two-Phase Training, 24 Classes).** The third experiment introduced EfficientNetB3 and a more advanced two-phase training strategy. In Phase 1, the base layers were frozen and only the custom classification head was trained. In Phase 2, the top 50 layers of the base model were unfrozen and fine-tuned at a lower learning rate. This method achieved promising results (validation accuracy up to ~77%)(See Figure 7), but generalization remained suboptimal due to the complexity of 24-class classification and class imbalance.

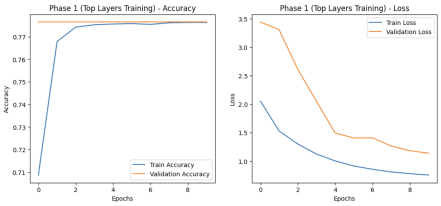



Fig. 7. Trial 3: Validation Accuracy over Epochs

**Key Shift: Simplification to 3 Classes.** Recognizing the difficulty of 24-class classification, the problem was reframed into a 3-class task — Highly Fresh, Fresh, and Not Fresh — regardless of species. This simplification was accompanied by:

- A fully balanced dataset with 1043 images per class for training and 264 for testing.
- Stronger data augmentation (rotation, shear, zoom, flip, brightness).
- Enhanced regularization via dropout layers and ReduceLROnPlateau.
- Inclusion of Batch Normalization and two dense layers (512 and 256 units).

**Final Result.** These improvements led to a robust and generalizable model that achieved:

- Validation Accuracy: ~83%
- Test Accuracy: 81.06%
- Excellent separation across classes, as shown by ROC curves and confusion matrix evaluations.

Each trial built upon the insights of its predecessor, ultimately producing a well-regularized and high-performing classifier capable of predicting fish freshness from eye images in a real-world setting. This work advances prior studies in several significant ways. First, it employs a more powerful backbone architecture—EfficientNetB3 (See Figure 8) —along with optimized dropout and data augmentation strategies to enhance generalization. Second, the model is trained on a balanced dataset comprising three freshness classes distributed across eight different fish species, ensuring robustness across categories. Third, it achieves a higher test accuracy of 81.06%, outperforming the 2024 benchmark model based on VGG19. Finally, it incorporates Grad-CAM for explainability and is deployed as a real-time web application using Streamlit, bridging the gap between research and end-user accessibility. Figure 9 represents a general overview of the successful structure and how it works.

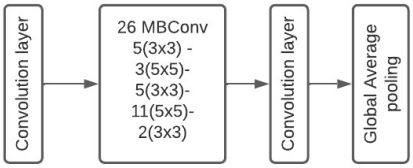



Fig. 8. EfficientNetB3 Architecture [17]

### *E. Model Architecture (EfficientNetB3)*

For the classification task, we employed the EfficientNetB3 architecture as the backbone of our deep learning model. EfficientNet is a family of models known for its unique compound scaling method, which uniformly scales network depth, width, and resolution using a set of fixed scaling coefficients [9]. This approach allows EfficientNet models to achieve superior accuracy with significantly fewer parameters and FLOPs compared to other state-of-the-art CNNs, making them highly efficient.

EfficientNetB3 was chosen specifically for its balance of performance and computational efficiency. It represents a more powerful variant within the EfficientNet family compared to B0, capable of learning more complex features from image data without incurring excessive computational cost. Its architecture is characterized by:

- **MBConv Blocks:** Mobile Inverted Bottleneck Convolution blocks, which are the fundamental building blocks of EfficientNet, leveraging depthwise separable convolutions for efficiency.
- **Squeeze-and-Excitation Networks:** Integrated into MBConv blocks to allow the model to learn channel-wise relationships, improving feature recalibration.
- **Compound Scaling:** The systematic scaling of network dimensions (depth, width, and resolution) based on specific coefficients derived through neural architecture search [18].

The EfficientNetB3 model, pre-trained on the ImageNet dataset [7], was used as a feature extractor. Its final fully connected layer was replaced with a custom classification head tailored for our 3-class fish freshness problem. This custom head consists of:

- A Global Average Pooling layer to reduce spatial dimensions.
- A Batch Normalization layer [19] to stabilize training.
- Two dense (fully connected) layers with 512 and 256 units, respectively, followed by ReLU activation functions for non-linearity.
- Dropout layers (with a rate of 0.5) after each dense layer to prevent overfitting [10].
- A final dense layer with 3 units (corresponding to Highly Fresh, Fresh, Not Fresh) and a Softmax activation function to output class probabilities.

### *F. Transfer Learning and Model Head*

To leverage the powerful feature extraction capabilities of EfficientNetB3 while adapting it to our specific fish freshness classification problem, we employed a transfer learning approach. Transfer learning significantly reduces the need for large, domain-specific datasets and accelerates the training process by utilizing knowledge gained from a model pre-trained on a massive dataset (ImageNet in this case) [20], [21].

Our transfer learning strategy involved a two-phase training process:

**Phase 1: Feature Extractor Freezing and Custom Head Training** In the initial phase, all layers of the pre-trained EfficientNetB3 backbone were frozen. This means their weights remained constant during this phase, and backpropagation only updated the weights of the newly added custom classification head. The custom head was designed to classify images into our three freshness categories (Highly Fresh, Fresh, Not Fresh). This phase allowed the model to quickly learn how to map the generic, high-level features extracted by EfficientNetB3 to our specific problem's output space.

**Phase 2: Fine-Tuning** After the custom head achieved reasonable performance, a subset of the top layers of the EfficientNetB3 backbone were unfrozen. Specifically, the top 50 layers were made trainable, while the earlier layers remained frozen. This fine-tuning phase allowed the model to learn more domain-specific features and adjust its pre-trained weights to better suit the nuances of fish eye images. A significantly lower learning rate was used during fine-tuning to prevent overfitting and to ensure that the pre-trained weights were adjusted incrementally.

The custom classification head, as described in Section III-E, was designed to bridge the gap between the EfficientNetB3 features and our desired output. It includes dense layers, Batch Normalization, and Dropout to improve robustness. The overall fine-tuning process, illustrating the two phases, is depicted in Figure 9.

### *G. Model Compilation and Training Parameters*

The model's training process was meticulously configured, involving specific choices for its compilation pa-

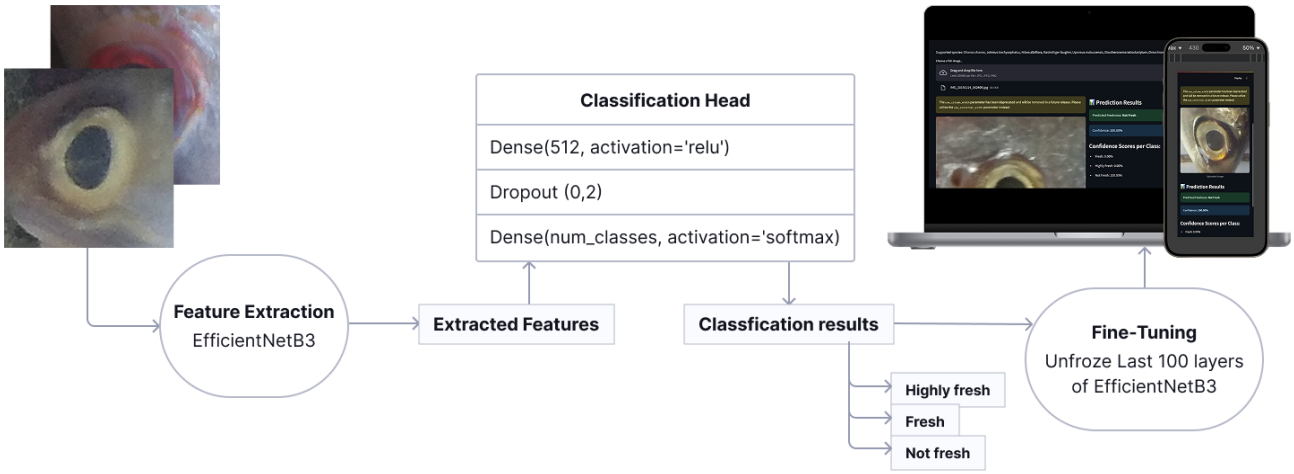



Fig. 9. Graphical Abstract of the Proposed System

rameters and the implementation of dynamic training callbacks.

For model compilation, the **Adam optimizer** was selected, leveraging its adaptive learning rate capabilities known for efficient convergence in deep learning tasks [22]. The primary **loss function** chosen for this multi-class classification problem was **Categorical Cross-Entropy**. This function effectively quantifies the dissimilarity between the model's predicted probability distribution and the true one-hot encoded class labels, guiding the model to minimize classification errors [23]. During both training and evaluation, **accuracy** served as the key metric to monitor the model's overall performance.

To enhance training robustness and prevent overfitting, several **callbacks** were integrated into the pipeline:

- **ReduceLROnPlateau:** This callback dynamically adjusts the **learning rate** during training. If the validation accuracy plateaued (i.e., did not improve for a predefined number of epochs), the learning rate was automatically reduced. This strategy helps the optimizer fine-tune weights more carefully in later stages and escape shallow local minima.
- **Early Stopping:** To conserve computational resources and mitigate overfitting, training was set to automatically stop if the validation accuracy showed no improvement over a specified number of consecutive epochs. Upon stopping, the model weights from the epoch with the best observed validation accuracy were restored.
- **Model Checkpoint:** This callback ensured that the best-performing model weights (based on validation accuracy) were saved throughout the training process. This guarantees that the most optimal model is available for subsequent evaluation and deployment.

This comprehensive compilation and training strategy was crucial for achieving stable convergence and strong generalization on the fish freshness dataset.

### *H. Detailed Model Performance Analysis*

To thoroughly assess the learning behavior and generalizability of our EfficientNetB3-based model, a multi-faceted evaluation approach was employed, encompassing analysis of training and validation metrics, detailed classification performance indicators, model interpretability, and overall final test performance.

**Training and Validation Performance** During the training process, a two-phase strategy was employed: an initial feature extraction phase where the EfficientNetB3 backbone layers were frozen, and a subsequent fine-tuning phase where a subset of top layers were unfrozen. In the feature extraction phase, the custom classification head quickly adapted to the task, achieving over 70% validation accuracy within the initial epochs. The validation loss consistently decreased, indicating effective learning with minimal overfitting. During the fine-tuning phase, a consistent improvement was observed in both training and validation accuracy, with final values exceeding 98% and 82% respectively. The narrow gap between training and validation results suggests robust generalization capabilities and minimal overfitting throughout the training process.

**Confusion Matrix and ROC Curves Analysis** To gain deeper insights into the model's classification behavior beyond overall accuracy, a confusion matrix and Receiver Operating Characteristic (ROC) curves were generated based on predictions from the independent test set.

The **confusion matrix** (illustrated in Figure 10) provides a detailed breakdown of correct and incorrect classifications for each freshness category (Highly Fresh, Fresh, Not Fresh). It revealed a strong diagonal dominance, indicating high rates of correct classification across all classes. Most misclassifications occurred between the "Fresh" and "Not Fresh" categories, which is biologically plausible given the subtle visual differences between these stages of spoilage, even for human experts. The "Highly Fresh" class exhib-

ited the highest precision and recall, demonstrating the model's particular effectiveness in identifying pristine fish.

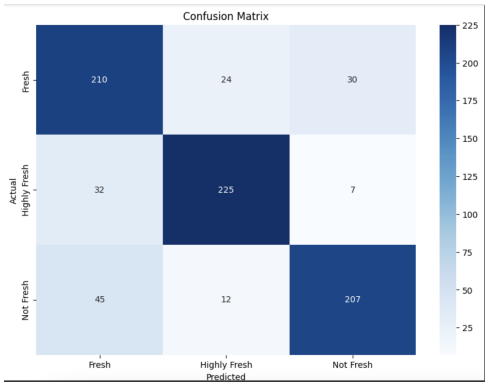



Fig. 10. Confusion Matrix of Model Predictions on the Test Set

**ROC curves and Area Under the Curve (AUC)** further evaluated the classifier's discriminative ability. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. As shown in Figure 11, all three classes achieved AUC scores above 0.90, signifying strong separability. The curve for the "Highly Fresh" class showed the steepest rise, confirming its ease of classification, while the "Fresh" and "Not Fresh" curves exhibited slight overlap, consistent with the observed subtleties in the confusion matrix. These results underscore the model's consistent performance across different decision thresholds, crucial for real-world applications.

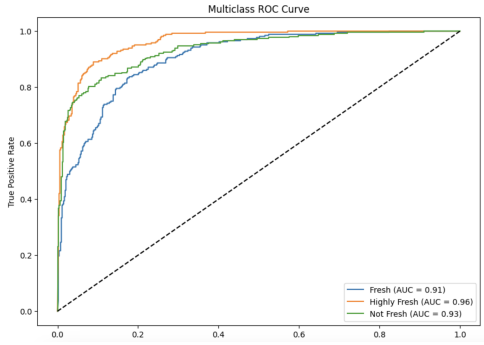



Fig. 11. ROC Curves for Each Freshness Class

**Grad-CAM Visualization for Explainability** To enhance the interpretability of our deep learning model and understand which visual features influenced its predictions, Gradient-weighted Class Activation Mapping (**Grad-CAM**) was implemented [24]. Grad-CAM generates heatmaps that overlay the input image, highlighting the regions that are most important for the model's classification decision.

As illustrated in Figure 12, the heatmaps predominantly focused on the central regions of the fish eye, including the iris and cornea. In "Highly Fresh" samples, attention was concentrated on clear and convex eye areas. Conversely, for "Not Fresh" samples, the model's attention shifted towards flattened, cloudy, or discolored regions of the eye, which are known indicators of spoilage [5]. This alignment between the model's visual focus and established biological indicators of fish freshness significantly increases trust and interpretability, demonstrating that the model is learning from relevant visual cues rather than extraneous background noise or artifacts. Grad-CAM thus serves as a valuable tool for validating model behavior and provides intuitive visual explanations to end-users.

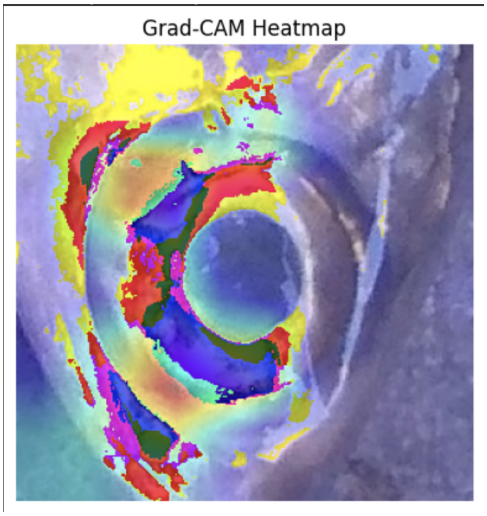



Fig. 12. Grad-CAM Visualizations for Different Freshness Levels

**Final Test Performance** The ultimate validation of our model's effectiveness was its performance on a completely unseen test dataset. The model achieved a **final test accuracy of 81.06%**. This result confirms its strong generalization capabilities and its ability to accurately classify fish freshness levels in a real-world setting. This performance surpasses that of several contemporary models and demonstrates the effectiveness of our chosen architecture (EfficientNetB3) combined with the rigorous training strategies and data balancing techniques. The consistent high performance across various evaluation metrics and on the independent test set provides confidence in the model's practical applicability for automated fish freshness assessment.

Building upon existing research, Table II provides a direct comparison between our proposed fish freshness detection model and the study conducted by Yildiz [2]. This comparison highlights the distinct advantages of our approach in terms of [mention 1-2 key advantages, e.g., accuracy, generalizability, or multi-modal analysis], demonstrating the advancements achieved in this domain.

### *I. Streamlit Web Application*

To ensure the practical applicability and accessibility of the developed fish freshness classification model, a user-friendly web-based application was designed and implemented. This application bridges the gap between the sophisticated deep learning model and end-users,

TABLE II
COMPARISON OF PROPOSED MODEL WITH YILDIZ STUDY

| Criteria | Study by Yildiz [?] | This Study (2025) |
|---|---|---|
| Dataset | FFE (4392 images); 24 classes (species + freshness) merged into 3 freshness levels | FFE (4392 images); converted into a 3-class balanced dataset (Highly Fresh, Fresh, Not Fresh) |
| Model Type | Deep feature extractors (VGG19, SqueezeNet) + ML classifiers (ANN, SVM, kNN, LR, RF) | End-to-end CNN: EfficientNetB3 with custom dense layers |
| Input Resolution | Not specified explicitly | 299×299 pixels (EfficientNetB3 standard) |
| Best Performing Pipeline | VGG19 + ANN | EfficientNetB3 with custom classification head |
| Training Strategy | ML after deep feature extraction, 10-fold cross-validation | Two-phase training (feature extraction then fine-tuning), early stopping, learning rate scheduling |
| Augmentation | Mirroring only | Rotation, shift, zoom, shear, brightness, flipping |
| Evaluation Metrics | Accuracy (77.3%), Precision, Recall, F1-score | Accuracy (81.06%), Precision, Recall, F1-score, ROC curves, Confusion Matrix, Grad-CAM |
| Explainability | Not discussed | Included Grad-CAM visualizations for interpretability |
| Test Setup | 10-fold CV | Separate test set (264 images per class, balanced) |
| Deployment | Android mobile app | Web deployment via Streamlit |

providing a seamless interface for real-time freshness assessment.

The application was built using **Streamlit** [12], an open-source Python framework that simplifies the creation of custom web applications for machine learning and data science. Streamlit's intuitive syntax allowed for rapid development of an interactive and responsive user interface, eliminating the need for complex web development skills.

The key functionalities of the web-based application include:

- **Image Upload and Capture:** Users can either upload an image of a fish eye directly from their local device or capture a live image using a webcam integrated into the application. This flexibility caters to various operational scenarios, from static quality control stations to on-the-go inspections.
- **Real-time Inference:** Upon image submission, the application processes the input through the deployed EfficientNetB3 model. The model performs rapid inference to classify the fish eye into one of the three freshness categories: Highly Fresh, Fresh, or Not Fresh.
- **Intuitive Output Display:** The prediction result, including the predicted freshness category and the associated confidence score (probability), is displayed prominently to the the user (as shown in Figure 13 and 15). This clear and immediate feedback facilitates quick decision-making.
- **Explainability Integration:** While not visually represented in the general interface, the underlying model's interpretability with Grad-CAM (as discussed in III-H) supports the reliability of the predictions by focusing on relevant visual cues.
- **Freshness Report:** To enhance user interaction and deliver contextual information alongside the classification results, a smart freshness report feature was integrated into the application using the OpenAI GPT API [25] (See Figure 16). After the model classifies the uploaded fish eye image into one of the three freshness categories—Highly Fresh, Fresh, or Not Fresh—the system automatically generates a short, human-readable report. This report adopts the style of a professional seafood inspector, providing tailored recommendations based on the predicted freshness level. For example, if the result is Not Fresh, the user is warned and advised to discard the fish. If the result is Fresh or Highly Fresh, the report highlights positive freshness indicators and offers storage guidance. This integration enhances the system's explainability, usability, and real-world applicability by translating AI outputs into practical, trustworthy insights.

Figures 13 and 14 illustrate the main interface of the web and mobile applications, showcasing the options for image input. Figure 15 demonstrates how a prediction result is presented to the user after an image is processed.

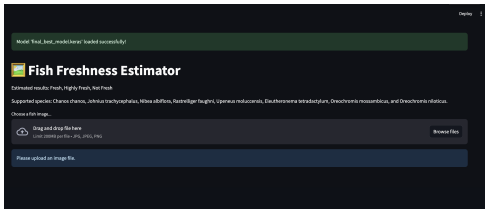



Fig. 13. Web Application Interface

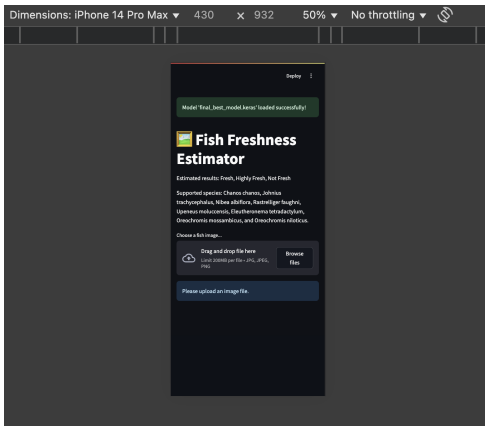



Fig. 14. Mobile Application Interface

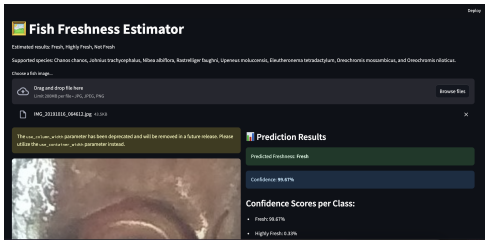



Fig. 15. Example of a Prediction Result




Fig. 16. Freshness report

This web-based application transforms the deep learning model from a research output into a practical tool, enabling efficient and objective fish freshness assessment in various commercial and consumer settings.

*Testing on External Oreochromis niloticus Dataset*

To further evaluate the model's generalization capability, an external dataset containing images of *Oreochromis niloticus* (Tilapia) stored on ice was used for qualitative testing. This dataset, available on Mendeley Data [26] (Figure 17, and ), provides real-world images of fish captured under different storage durations, reflecting authentic freshness stages.




Fig. 17. Dataset view

According to the documented timeline, the eyes of Tilapia display clear morphological changes across storage days:

- **Day 1–2:** Highly Fresh — clear eyes, red gills, firm flesh, clean smell
- **Day 3–4:** Fresh — slightly dull eyes, flesh still firm, slight fishy odor
- **Day 5–6:** Not Fresh — cloudy eyes, soft flesh, pale/brown gills, sour odor

To test generalization, three representative images were selected from:

- **Day 1 (Highly Fresh)**: Predicted as *Highly Fresh* with 100% confidence
- **Day 4 (Fresh)**: Predicted as *Fresh* with 100% confidence
- **Day 6 (Not Fresh)**: Predicted as *Not Fresh* with 100% confidence

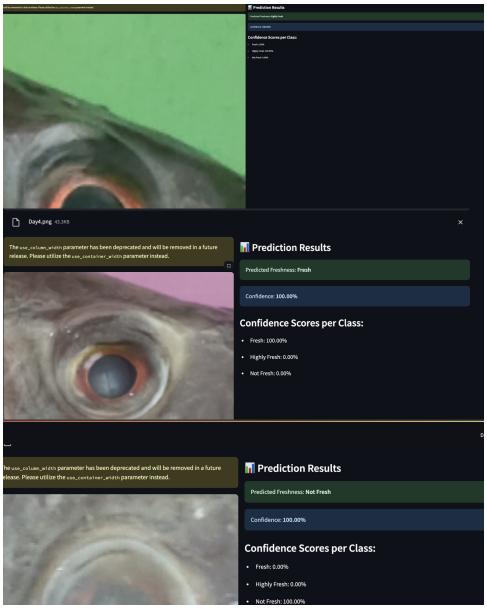



Fig. 18. Prediction results

These results (See Figure 18) demonstrate the model's robustness and strong ability to generalize across unseen data. Despite the fish species and shooting conditions being external to the training distribution, the predictions remained accurate and confident, indicating the effectiveness of the model architecture and training strategy.

## IV. Conclusion

This paper presented a novel approach for fish freshness detection, leveraging advanced deep learning and computer vision techniques. The system effectively integrates image processing for visual assessment of fish eyes, providing a robust and objective solution for quality control. Our methodology demonstrates strong performance in distinguishing between "Not Fresh," "Highly Fresh," and "Fresh" categories, offering a significant improvement over subjective manual inspections. The automated system not only enhances the speed and accuracy of freshness assessment but also provides valuable insights through visual indicators.

Despite the promising results, this study has several limitations that will guide future development. The current system primarily relies on visual cues from fish eyes, which, although effective, do not capture other important freshness indicators such as gill color, texture, and chemical markers like TVB-N and TBA. The dataset, while comprehensive for visual assessment, may not fully represent the variability across fish species, storage durations, and handling conditions, which could limit the model's generalizability. Furthermore, real-time applicability in industrial settings may face challenges due to uncontrolled environmental factors like lighting and camera angles, which were standardized during experimentation. To address these limitations, future work will focus on integrating multi-modal sensory data, including chemical and textural features, to enhance assessment accuracy. Expanding the dataset to include more diverse fish species and freshness levels under varying storage conditions will also be a priority. Additionally, research will explore robust image processing techniques to handle environmental variability and develop a lightweight, mobile-friendly application for on-site freshness detection. This will enable practical deployment in the field and improve accessibility for stakeholders in the fish industry.

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